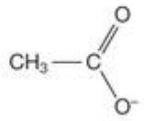
contains magnesium ions, Mg^{2+}, and ethanoate ions, CH_3COO^-. That means that you need two ethanoate ions for every magnesium ion to make the charges balance. The metal is written after the ethanoate because that reflects the structure; the magnesium ion is attracted to the negative charge on the oxygen atoms in the ethanoate ions.



▲ Figure 27.4 The structure of an ethanoate ion

Other metals behave as you would expect. Sodium is dangerously reactive; zinc produces bubbles of hydrogen more slowly than it does with hydrochloric acid, for example; metals below hydrogen in the reactivity series won't react.

Wherever a reaction takes place, a metal ethanoate will be formed as well as hydrogen. Zinc ethanoate is $(CH_3COO)_2Zn$.

EXTENSION WORK

Ethanoic acid is a weak acid, whereas the acids we have met before, hydrochloric acid or sulfuric acid, are called strong acids. Students can have the misconception that strong acids are equivalent to concentrated acids. Whether an acid is strong or weak has, in fact, nothing to do with its concentration. Strong acids dissociate completely into their constituent ions in water, for example in hydrochloric acid you find mostly H^+ and Cl^- ions and hardly any HCl molecules. In weak acids, the dissociation is only partial. About 1 in ~75 CH_3COOH molecules will dissociate into H^+ and CH_3COO^- in water. This is why if you have the same concentration of ethanoic acid and hydrochloric acid, the ethanoic acid reacts with metals or metal carbonates much more slowly than the hydrochloric acid as the concentration of the H^+ ions is much less.

REACTION WITH CARBONATES

Carbonates react with acids to give a salt, carbon dioxide and water (Chapter 17, page 177), and ethanoic acid behaves like any other acid. For example, with sodium carbonate you get a lot of fizzing and a colourless solution of sodium ethanoate:

$$Na_2CO_3(s) + 2CH_3COOH(aq) \rightarrow 2CH_3COONa(aq) + CO_2(g) + H_2O(l)$$

With calcium carbonate you again get lots of fizzing and a colourless solution – this time containing calcium ethanoate:

$$2CH_3COOH(aq) + CaCO_3(s) \rightarrow (CH_3COO)_2Ca(aq) + CO_2(g) + H_2O(l)$$

CHAPTER QUESTIONS

SKILLS INTERPRETATION

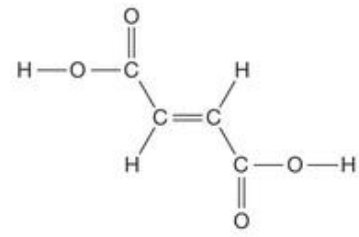
1 a Draw the structural formulae for the following compounds:
   i ethanoic acid
   ii butanoic acid

   b Draw the displayed formulae for these compounds:
   i methanoic acid
   ii propanoic acid

   c Name the following compounds:
   i $CH_3CH_2CH_2COOH$
   ii CH_3COOH

SKILLS CRITICAL THINKING (6)

2 What are the functional groups shown in the compound below (choose one answer from the table below)?



	Alkene	Alcohol	Carboxylic acid
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✗	✓	✓

SKILLS INTERPRETATION (8)

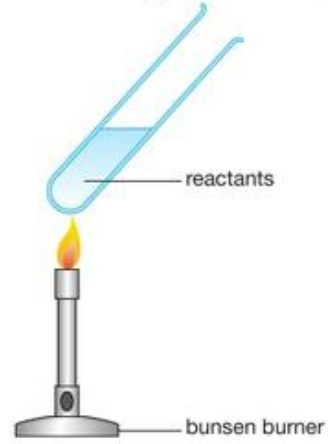
3 Propan-1-ol can be oxidised by heating it under reflux with potassium dichromate(VI) in dilute sulfuric acid.

a Draw the full displayed formula of the product of this reaction.

SKILLS CRITICAL THINKING
SKILLS EXECUTIVE FUNCTION (5)

b Describe the colour change during the reaction.

c A student suggested the following apparatus for the oxidation of propan-1-ol. Explain why this method might be too dangerous.



SKILLS CRITICAL THINKING, PROBLEM SOLVING (9)

d Describe what you would see and write balanced chemical equations for the reactions if you added some of the product you collected to test-tubes containing:

i magnesium ribbon ii sodium carbonate solution.

SKILLS PROBLEM SOLVING (8)

4 Vinegar can react with baking soda, which contains sodium hydrogencarbonate ($NaHCO_3$), to give a salt, carbon dioxide and water.

a Balance the following equation for the reaction between vinegar and baking soda.

$$\underset{\text{vinegar}}{_____} + \underset{\text{baking soda}}{NaHCO_3} \rightarrow \underset{\text{a salt}}{_____} + CO_2 + H_2O$$

SKILLS CRITICAL THINKING (6)

b Suggest the type of reaction occurring.

c Suggest an observation that could be made during this reaction.

END OF CHEMISTRY ONLY